



**NATIONAL COMPETENCY STANDARDS &
COMPETENCY BASED TRAINING CURRICULUM
FOR
COMPUTER HARDWARE TECHNICIAN
(Competency Standard Code: K72S001)**



**Developed and validated by;
National Apprentice & Industrial Training Authority
971, Sri Jayawardenepura Mawatha,
Welikada,
Rajagiriya.**



**Endorsed by;
Tertiary & Vocational Education Commission
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Ministry of Education, Higher Education and Vocational Education

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PREFACE

National Competency Standards (NCS) and curricula are essential elements for the implementation of a unified Technical and Vocational Education and Training System for Sri Lanka and forms the basis for the National Vocational Qualifications Framework of Sri Lanka, which provides for the award of national and internationally recognized qualifications. The National Competency Standards are developed in consultation with the industry and are designed using a nationally agreed specific format to maintain uniformity and consistency of standards amongst occupations.

The NCSs and curricula specify a competent worker's performance standards and the various contexts in which work may take place. NCS describes the knowledge, skills and attitudes required in a particular occupation and signals what students must be able to do in employment and can ideally serve as a link between education and training and the needs of the labour market. NCS and curricula together provide explicit advice to assessors and employers regarding the knowledge, skills and attitudes to be demonstrated by the candidates seeking formal recognition for the competencies acquired either following training or through work experience.

National Competency Standards focus on what is expected of a worker in the workplace rather than on the learning process.

- NCSs stress upon the ability to transfer and apply knowledge and skills to new situations and environments.
- In NCS and curricula the emphasis is upon outcomes and upon the application of knowledge and skills, not just the specification of knowledge and skills.
- NCS and curricula are concerned with what people are able to do and the ability to do a task in a range of contexts.
- NCS and curricula include all aspects of workplace performance and not only narrow task skills.

NCS especially can serve a number of purposes including;

- Providing advice to curriculum developers about the knowledge, skills and attitudes to be included in the curriculum.
- Providing specifications to Competency Based Assessors about the knowledge, skills and attitudes to be demonstrated by candidates.
- Providing advice to industry about job functions, which in turn can be used for the development of job descriptions, performance appraisal systems and workflow analysis. The lead organization for the development of NCS and curricula is the National Apprentice & Industrial Training Authority. The standards so developed are endorsed by the Tertiary & Vocational Education Commission as National Document.

There is a requirement to review the standards and curricula within the prescribed period as appropriate as and when required, with the assistance of relevant industry groups and incorporate the changes in the National Competency Standards and Curricula.

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NATIONAL COMPETENCY STANDARDS & COMPETENCY BASED TRAINING

CURRICULUM

FOR

COMPUTER HARDWARE TECHNICIAN

(Competency Standard Code: K72S001)

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Occupation Outlook

Job Description

Computer hardware technician prepare quotations, estimates, assemble computer systems, perform, installation and configuration of computer software, troubleshoot computer systems and peripheral devices, protect computer system from cyber-attacks, configure and maintain computer networks and carry out the preventive maintenance of the computer systems in Small Office, Home Office (SO/HO).

Responsibilities

- Prepare estimates/quotations
- Procure required components
- Assemble the computer
- Install, configure and update software
- Troubleshoot issues of computer systems and computer networks
- Fulfill customer care and service needs at the reception counter
- Perform tests to assess the maintenance and repair requirement
- Carryout repairing and upgrading task
- Carryout preventive maintenance
- Maintain a safe work environment and proper disposal of electronic waste

Career and Employment Opportunities

Trainees who have completed this competency level, can gain employment as computer hardware technicians and continue into higher NVQ levels and work as follows, but are not limited to:

- Customer help desk officer for computer hardware
- Preventive maintenance officer for computers / Networks
- IT support technician/ Technical officer
- Customer support/ Sales executive – Computer Hardware
- Computer hardware demonstrator
- Computer Lab Assistant

**NATIONAL COMPETENCY STANDARDS
FOR THE OCCUPATION OF
COMPUTER HARDWARE TECHNICIAN
(Competency Standard Code: K72S001)**

	List of Competency Units	Unit Code No.	Level	Page no:
1.	Develop estimates and quotations	K72S001U01	04	01
2.	Rectify issues of existing system	K72S001U02	04	04
3.	Customize the computer system configuration	K72S001U03	03	07
4.	Perform preventive maintenance of the computer system	K72S001U04	03	10
5.	Install, configure and update computer software	K72S001U05	03	13
6.	Upgrade hardware components and external devices	K72S001U06	03	16
7.	Handle computer network setup and configuration	K72S001U07	04	19
8.	Practice workplace communication and interpersonal relations	K72S001BU01	Basic	22
9.	Apply occupational literacy and numeracy	K72S001BU02	Basic	24
10.	Work in teams	K72S001BU03	Basic	26
11.	Practice occupational health and safety procedures in a workplace	K72S001BU04	Basic	28

Packaging for National Vocational Qualifications (NVQ)

- National certificate at **NVQ Level 3** in the occupation of **Assistant Computer Hardware Technician** will be awarded to those who are competent in U03+ U04+ U05+ U06+ BU01+ BU02+ BU03+ BU04 – **K72S001Q1L3**
- National certificate at **NVQ Level 4** in the occupation of **Computer Hardware Technician** will be awarded to those who are competent in U01+ U02+ U03+ U04 + U05+ U06 + U07+ BU01+ BU02+ BU03+ BU04 - **K72S001Q2L4**

Unit Title:	Develop estimates and quotations
Unit Descriptor:	This unit covers the competencies required to prepare estimates for repair, upgrade, new installation and preventive maintenance of computer systems adhering to the current market conditions and terms and conditions prevailing in the organization
Unit Code:	K72S001U01
Elements Of Competence	Performance Criteria
1. Gather customer requirement	1.1 <i>Customer requirement</i> obtained as per the SOP 1.2 <i>Details of the solution</i> collected according to market conditions 1.3 Solution/s suggested and agreed with the customer
2. Generate estimate/quotation	2.1 <i>Cost elements</i> selected for the agreed solutions as per the SOP 2.2 Estimate/quotation prepared for the agreed solution/s-according to the identified cost elements following standard formats 2.3 Approval of the prepared estimate/quotation obtained by authorized person 2.4 Prepared estimate/quotation submitted to the customer as per the SOP 2.5 Customer's approval obtained for submitted estimate/quotation as per the SOP 2.6 Records kept on approved estimate/quotation as per the SOP

Range Statement:

Work connected to this unit may take place at an office, workshop or any other suitable facility where the computer equipment could be safely used. Work related to this unit may be performed individually or as a member of team.

Computer repair/update areas are;

- Desktop Computers
- Laptop Computers
- Hardware components such as processors, cooling devices, SSD, etc.
- Peripherals such as printers, Scanners, etc.
- Other hardware devices such as UPS, Routers, Switches, etc.
- Computer software

New installation areas are;

- Computer system and other peripherals
- Configuration of wired and wireless network

Maintenance areas are;

- Preventive maintenance agreements

Elaboration of Terms in Performance Criteria

- ***Customer requirement*** in PC 1.1 may include: requirement of new installation, fault in the computer and peripheral devices, upgrading hardware & software, Preventive maintenance, etc.
- ***Details of the solution*** in PC 1.2 may include: present technology, current price, warranty conditions, availability, compatibility, delivery period & conditions, etc.

- **Cost elements** in PC 2.1 may include: Material cost, Labour cost, Transportation cost, Professional fee, Licencing cost, Taxes, Utility fee, Inspection charges, Overheads charges, etc.

Critical aspects

The assessment must confirm that the candidate is able to;

- Collect precise customer requirement and definite fault
- Interpret manufacturing warranty and terms & conditions

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
• External storage devices • Printer • Scanner • Laptop/Desktop computer/ Smart phone • Internet facility	• Stationary • Standard formats of estimate/quotation

The following documents /References/Standards may be used for this unit:

- Estimation guidelines of the company if available
- Technical specification / Data sheet
- Standard Operating Procedure (SOP)
- Standard Organization's procedures
- Manufacturing warranty and terms and conditions
- Government procurement guidelines
- Government notification on taxes

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Computer systems • Computer software applications and operating system • Computer hardware principles • Computer peripherals • Basics of-electricity and electronics. • Basics of computer network • Cost elements • Existing market prices ofcomputers, peripherals and other accessories • Procurement process • Government taxes • Arithmetic calculations	• Operate computer • Read and interpret manuals and specifications • Uses of Spread sheet and word processing • Dictating capability • Communication skill • Record keeping • Use of Internet and social media

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Integrity and honesty • Punctual and efficient • Accuracy • Helpfulness • Time for details • Team work • Knowledge seeking • Self-learning • Work under pressure

Unit Title:	Rectify issues of existing system
Unit Descriptor:	This unit covers the competencies required to diagnose the fault/s of computer systems and carrying out the hardware/ software/ network repair work and ensure the protection of computer system from cyber-attack while adhering to the safety and environmental regulations
Unit Code:	K72S001U02
Elements Of Competence	Performance Criteria
1. Ensure the validity of warranty	1.1 Warranty type of the computer system checked as per the SOP 1.2 Condition/Validity of the warranty seal checked and ensured before commencing the work, if it is within the warranty period of the same company 1.3 Customer/computer system directed to the relevant company, if it is within the warranty period of another company 1.4 Customer informed about the warranty terms and conditions as per the SOP
2. Gather customer requirement	2.1 Customer details and visual inspections recorded in the job card 2.2 Details of the hardware/network/software application environment of the customer collected as per the SOP 2.3 Information obtained from the customer regarding the fault/s in the computer systems 2.4 Job card completed as per the Standard Operating Procedure
3. Perform fault finding	3.1 Pre-preparation activities carried out as per the SOP 3.2 Customer's concurrence obtained for the detailed diagnosis of the computer system. 3.3 Computer system inspected to localize the faulty area and updated the job card. 3.4 Specific software/hardware/network fault of the inspected computer system and related peripherals diagnosed as per the SOP
4. Rectify the fault	4.1 Required hardware/network /software repair tools selected and arranged as per the diagnosed fault/s 4.2 Hardware/network components cleaned using cleaning agents as per the manufacturer's guidelines 4.3 Replaceable hardware /network /software components obtained checking the compatibility as per SOP, if required 4.4 Repairable hardware /network /software components attended as per the guidelines given by component manufacturer, if required 4.5 System operation checked to ensure the smooth function of the computer system 4.6 Waste disposed as per the environmental regulations

5. Complete the repair task	5.1 Rectification work reviewed with the customer and customer acceptance obtained as per the SOP 5.2 Job card updated as per the SOP 5.3 Invoice/s and payment procedure followed as per the SOP and approved estimate/quotation
6. Ensure the protection of computer system from cyber attack	6.1 Activation/updating of antivirus checked and ensured as per SOP 6.2 Effect of malicious software identified by scanning using relevant software 6.3 Identified issues related to infections fixed using appropriate tools 6.4 Functioning of computer software and devices rechecked as per SOP 6.5 Customer advised on protecting the computer system from cyber attack

Range Statement:

Work connected to this unit may take place at an office, workshop or any other suitable facility where the computer equipment could be safely used. Work related to this unit may be performed individually or as a member of team.

Elaboration of Terms in Performance Criteria

- **Warranty type** in PC 1.1 may include: Comprehensive, Service
- **visual inspections** in PC 2.1 may include: physical damages, accessories, missing parts, etc.
- **Pre-preparation activities** in PC 3.1 may include: PPE/static protective cloths worn, maintain personal hygiene, maintain professional image, etc.
- **appropriate tools** in PC 6.3 may include: virus guard software, cleaning tools, specific cleaners

Critical aspects

The assessment must confirm that the candidate is able to;

- Obtain information from the customer regarding the fault/s in the computer systems correctively
- Obtain requirements for rectify fault
- Carry out required corrective action

The following tools, equipment & material are included within this unit

Tools and equipment	Materials
• Computer diagnostic tools (Hardware & Software) • Computer Hardware toolbox • Anti-malware software • External storage devices • Booting devices • Network toolbox • PPE • Vacuum/Blower • Cleaning brushes	• Stationery • Cleaning agents

The following documents /References/Standards may be used for this unit:

- Software License
- Software manuals
- Device manufacturers' guidelines
- Technical data sheet
- Material Safety Data Sheet (MSDS)
- Standard Operating Procedure (SOP)
- Manufacturing warranty and terms and conditions
- Occupational Health and Safety guidelines
- Environmental protection guidelines
- Cyber security guidelines

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Computer systems • Computer software applications and operating systems • Computer hardware principles and technical specification • Computer peripherals & accessories • Basics of electricity and electronics. • Basics of networks • Basic knowledge on cyber security • Antivirus software and tools • New technologies, devices and applications • Existing market prices of computers peripherals and other accessories • Artificial Intelligent (AI) • Occupational Health and Safety procedures	• Read and interpret manuals and specifications • Computer operate and use of application software • Handling tools and diagnostic software • Performing to meet dead lines • Selection and application of proper computer hardware • Selection and application of suitable software and tools for diagnosis. • Use of antivirus software and tools

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Politeness • loyalty • Integrity and honesty • Punctual and efficient

Unit Title:	Customize the computer system configuration
Unit Descriptor:	This unit covers the competencies required to assemble a computer system and test the computer system while adhering to the health, safety and environmental regulations
Unit Code:	K72S001U03
Elements Of Competence	PERFORMANCE CRITERIA
1. Prepare job card	1.1 <i>Details of the customer</i> filled in the job card as per the SOP 1.2 Job card updated with the customer requirement and estimate/quotation
2. Obtain required components	2.1 Required components collected/purchased as per the job card 2.2 Collected/purchased components verified with the order/job card
3. Initiate assembling process	3.1 Personal Protective Equipment (PPE) worn as per SOP 3.2 Required tools & equipment gathered as per the requirement 3.3 Workplace arranged as per the requirement while adhering health and safety guidelines
4. Complete assembling process	4.1 Components assembled following the component manufacturer guidelines 4.2 Completeness of assembling order/sequence of the assembled components ensured as per manufacturer guidelines 4.3 Power connected to the assembled computer system and initialized as per SOP 4.4 System functions tested using relevant diagnostic tools
5. Carry out post assembling tasks	5.1 Assembled computer system forwarded for computer software installation as per the SOP 5.2 Used tools and equipment cleaned and stored as per the SOP 5.3 Work bench rearranged as per SOP 5.4 Actions taken to disposed E–waste under supervision of the superior adhering to environmental regulations

Range Statement:

This unit may take place at workshop/ work environment. Work related to this unit may be performed individually or as a member of team.

Customize areas are;

- Desktop computer/ Workstation/gaming PC
- Laptop

Elaboration of Terms in Performance Criteria

- *Details of the customer* in PC 1.1 may include: Customer name, contact details (Postal Address, Email address, Telephone no, site location)

Critical aspects

The assessment must confirm that the candidate is able to;

- Use relevant tools
- Follow correct assembling sequence
- Assemble components without any damages
- Adhere to the environmental protection regulation and OHS guidelines

The following tools, equipment & material are included within this unit

Tools and equipment	Materials
• Computer Hardware tool box • Diagnostic tools • Bootable devices • PPEs	• Cleaning material • Stationary

The following documents /References/Standards may be used for this unit:

- Device manufacturers' guidelines
- Technical data sheet
- Standard Operating Procedure (SOP)
- Manufacturing warranty and terms and conditions
- Occupational Health and Safety guidelines
- Manufacturer guidelines
- Central environmental authority regulations

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Computer components • Testing methods & procedures • Assembling order/sequence • Standard operating procedures • Diagnosing /testing tools • Basics of Electricity and Electronics • PPEs • Environmental regulation • Error detection beep codes and correction methods • CMOS (BIOS /UEFI) settings	• Communication and presentation • Computer operations • Read and interpret manuals • Handle hardware components • Use of computer diagnostic tools • Use of Personal Protective Equipment

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Integrity and honesty • Punctual and efficient • Team work • Attention to the details • Ergonomics

Unit Title:	Perform preventive maintenance of the computer system
Unit Descriptor:	This unit covers the competencies required to review the maintenance agreement, carry out preventive maintenance or services on computer system, while adhering to the safety, health and environmental regulations
Unit Code:	K72S001U04
Elements Of Competence	Performance Criteria
1. Initiate the preventive maintenance work.	1.1 Preventive maintenance agreement reviewed as per the SOP 1.2 Required tools and equipment gathered as per the SOP 1.3 Computer system verified as per the preventive maintenance agreement 1.4 Current condition of the computer system discussed with the user as per the SOP 1.5 Current condition of the computer system checked and documented as per the SOP 1.6 Restore point created following standard procedures 1.7 Backing up carried out on the selected storage media if required
2. Carry out hardware maintenance work	2.1 Computer system de-assembled following the SOP 2.2 Static charges discharged following the SOP 2.3 Dust removed following the SOP 2.4 Ports and connectors de-oxidized using contact cleaners as per the SOP 2.5 Thermal compound replaced following the SOP 2.6 System case cleaned following the SOP 2.7 Computer system assembled and tested for proper functioning as per the equipment manufacturer's guidelines 2.8 Backup restored as per the requirement 2.9 Actions taken to disposed E-waste under supervision of the superior adhering to environmental regulations
3. Carry out software maintenance work	3.1 Software issue/s verified according to the condition of the system and documented 3.2 Operating system updated following the SOP if required 3.3 Temporary files cleaned following SOP 3.4 Disk cleanup performed following SOP 3.5 Unnecessary/conflicting application software uninstalled with the user consent 3.6 Device drivers updated following the standard procedures if required

	3.7 Anti-malware and <i>other required utilities</i> updated following the standard procedures if required
4. Instruct customer on trouble free operations	4.1 Current situation of Hardware/Software of the computer system informed to customer as per the SOP 4.2 Anticipated failures/Replacements of Hardware/Software of the computer system documented and informed to the customer as per the SOP 4.3 <i>Required settings of the computer system</i> described and acknowledged

Range Statement:

This unit may take place at workshop/ work environment. Work related to this unit may be performed individually or as a member of team.

Elaboration of Terms in Performance Criteria

other required utilities in PC 3.7 may include software tools used to enhance user functions. chatting software, backup tools, etc.

Required settings of the computer system in PC 4.3 may include: making restore point, getting proper backup, etc.

Critical aspects

The assessment must confirm that the candidate is able to;

- Check the availability of the maintenance agreement/ job card and interpret accurately
- Obtain user/customer consent to carry out the job
- Check and verify the requirement to backup data/ software

The following tools, equipment & material are included within this unit.

Tools & equipment	Materials
• PC tool box • Vacuum/Blower • Diagnostic tools • Smart phone • PPE	• Heat sink compound • Cleaning agents • Cleaning material • Lubricants • Stationary

The following documents /references/standards may be used for this unit:

- Maintenance agreement/ job card
- Device/Equipment manufacturer guidelines
- Technical data sheet
- Standard Operating Procedure (SOP)
- Manufacturing warranty and terms and conditions
- Material Safety Data Sheet (MSDS)
- Occupational Health and Safety guidelines
- Environmental protection guidelines
- Cyber security guidelines

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Computer components • Testing methods & procedures • Disassembling and Assembling order/sequence/procedures • Elementary of electrical/electronic concepts • Standard Operating Procedures • Diagnosing /testing tools • PPE • Environmental regulation	• Communication and presentation • Computer operations • Read and interpret manuals/agreement • Identification of computer components • Handling of Hardware and software • Use of testing equipment (multimeter, DC power supply) • Use of computer diagnostic tools • Use of Personal Protective Equipment • Safe handling of chemicals

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Integrity and honesty • Punctual and efficient • Team work • Attention to the details • Ergonomics

Unit Title:	Install, configure and update computer software
Unit Descriptor:	This unit covers the competencies required to install, configure and update selected operating systems, drivers, application software and utilities while adhering to the safety and environmental regulations.
Unit Code:	K72S001U05
Elements Of Competence	Performance Criteria
1. Install and configure operating system	1.1 Operating system selected as per the customer requirement 1.2 Hard Disk Drive (HDD)/Solid State Drive (SSD) partitioned as per the customer requirement 1.3 Created partitions formatted as required 1.4 Installation of the selected operating system, performed as per operating system guidelines 1.5 Activation of the installed operating system performed as per the operating system guidelines 1.6 Required device drivers installed/updated as per system hardware 1.7 Configuration of operating system customized as per customer requirement 1.8 Functions tested and verified for trouble free operation
2. Install application software	2.1 Application software selected as per the customer requirement 2.2 Compatibility of hardware & software verified for the selected application software 2.3 Infrastructure/ software customized as necessary 2.4 Installation of the selected application software performed as per SOP 2.5 Functions tested and verified for trouble free operation
3. Install and configure utilities	3.1 Utility software selected as per the customer requirement 3.2 Compatibility of hardware & software verified for the selected utility software 3.3 Installation of the selected utility software, performed as per utility software guidelines 3.4 Configuration of utility software customized as per customer requirement 3.5 Functions tested and verified for trouble free operation
4. Update software	4.1 Software update identified as per the system requirement 4.2 Update of the selected software performed as per relevant guidelines 4.3 Function tested and verified for trouble free operation 4.4 Customer/User made aware on updating procedures as per the SOP

Range Statement:

Work connected to this unit may take place at an office, workshop or any other appropriate location where limited Local Area Network (LAN) could be safely handled. Work related to this unit may be performed individually or as a member of team.

Elaboration of Terms in Performance Criteria

- **External devices in PC 4.1** may include: biometric devices using attendance and access control systems, network printers, surveillance system, cloud storage etc

Critical aspects

The assessment must confirm that the candidate is able to;

- Verify the functionality of network connectivity
- Verify the functionality of the internet & E- mail
- Verify the functionality of the external devices connected to the network

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
Computer system	Manuals
Operating system with bootable devices	Stationary
Device drivers	Network/Patch cables
Network devices	RJ-45 Jacks
Utility software	
Internet facility	
Application software	
PC repair tool box	
Network tool box	
Cable tester and probe	
Multimeter	
Drill	
PPE	

The following documents /References/Standards may be used for this unit:

- Software License
- Device manufacturers' guidelines
- Technical data sheet
- Standard Operating Procedure (SOP)
- Manufacturing warranty and terms and conditions
- Occupational Health and Safety guidelines

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Legal aspects of the computer software • Operating systems • Application software • Peripheral devices • Structured Cabling • Cabling standards • Fundamentals of networking • Functions of network devices • Communication techniques (guided (wired)/ unguided(wireless)) • Utility software • Ports, adapters and connectors • External devices which could be connected to computer network • Drivers and their application • Information security and safety • Operating system migration • System Cloning • Ethical use of computer software • Waste disposal methods	• Communication • Team work • Read and interpret manuals • Record keeping skills • Use of internet • Use of PPE • Cable laying and jointing

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Politeness • loyalty • Integrity and honesty • Punctual and efficient

COMPETENCY UNIT:	Upgrade Hardware Components and External Devices
UNIT DESCRIPTOR:	This unit covers the competencies required to identify upgrade and install of hardware components and external devices and perform the upgrade and installation while adhering to safety, health and environmental guidelines.
UNIT CODE:	K72S001U06
ELEMENTS OF COMPETENCE	PERFORMANCE CRITERIA
1. Install hardware components	1.1 Hardware components identified for upgrading as per the customer requirement 1.2 System capability checked and verified for upgrading according to existing system/customer requirement 1.3 Required hardware components listed out and obtained as per the SOP 1.4 Obtained hardware components fixed according to the manufacturer's guidelines 1.5 Function tested and verified for trouble free operation 1.6 Documentation completed as per the SOP 1.7 Actions taken to disposed E-waste under supervision of the superior adhering to environmental regulations
2. Setup external devices	2.1 External devices/cloud storage identified as per the customer requirement/environment 2.2 Compatibility of hardware / software of the external device/cloud storage verified according to available system 2.3 External device/cloud storage and device drivers/software installed/configured as per the manufacturer's guidelines 2.4 Functionality of the Installed device tested and commissioned as per the customer requirement 2.5 Documentation completed as per the SOP 2.6 Work environment cleaned following the disposal guidelines

Range Statement:

Work connected to this unit may take place at an office, workshop or any other appropriate location where the computer system could be safely handled. Work related to this unit may be performed individually or as a member of team.

Elaboration of Terms in Performance Criteria

- **System capability in PC 1.2** may include: port, slot, capacity, version, device bay, etc.

Critical aspects

The assessment must confirm that the candidate is able to;

- Verify the compatibility of the hardware components with the existing system
- Ensure the functionality of the installed hardware components
- Ensure the functionality of the installed external devices

The following tools, equipment & material are included within this unit

Tools and equipment	Materials
• Computer hardware components • External devices • Computer system • Operating system with bootable devices • Device drivers • Utility software • Internet facility • Application software • Computer hardware tool box • PPE	• Manuals • Stationary

The following documents /References/Standards may be used for this unit:

- Software License
- Device manufacturers' guidelines
- Software developers' guidelines
- Technical data sheet
- Standard Operating Procedure (SOP)
- Manufacturing warranty and terms and conditions
- Occupational Health and Safety guidelines
- Central Environmental Authority (CEA) rules and regulations related to industry (National Environment Act No.47 of 1980)

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Legal aspects of the computer software • Operating systems • Application software • Utility software • Peripheral devices • Ports, adapters and connectors • Drivers and their application • Information security and safety • Operating system migration • System Cloning • Ethical use of computer software • E-Waste disposal methods • PPE	• Communication • Team work • Read and interpret manuals • Record keeping skills • Use of internet • Use of PPE

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Politeness • loyalty • Integrity and honesty • Punctual and efficient

COMPETENCY UNIT:	Handle computer network setup and configuration
UNIT DESCRIPTOR:	This unit covers the competencies required to setup/ install network, external devices and configure the network for communication functions while adhering to the security and safety guidelines.
UNIT CODE:	K72S001U07
ELEMENTS OF COMPETENCE	PERFORMANCE CRITERIA
1. Initiate the network requirement	1.1 New/Upgrade computer network requirement identified as per the customer requirement 1.2 Solution identified and options discussed with the customer 1.3 Network layout and power supply layout sketched as per the customer requirement 1.4 Approval obtained from the customer to the finalized network layout 1.5 Required items procured as per the approved quotation by the customer
2. Setup / Install network	2.1 Physical connectivity performed as per the approved network layout 2.2 Required software installed /updated as per the SOP 2.3 Computer network configured as per the customer requirement 2.4 Functionality of the network tested and verified for trouble free operation 2.5 Documentation completed as per the SOP 2.6 Work environment cleaned following the disposal guidelines
3. Setup connectivity for communication	3.1 Internet connectivity configured as per the customer requirement 3.2 Email account/s configured as per the customer requirement 3.3 Chat tool/s installed and configured as per the customer requirement 3.4 Social network settings configured as per the SOP 3.5 Documentation completed as per the SOP
4. Connect external devices to the network	4.1 Connected/to be connected External devices identified as per customer requirement 4.2 External devices installed and configured according to manufacturer's guidelines 4.3 Configured external devices tested as per manufacturer's guidelines 4.4 Customer instructed to the proper use of connected external devices
5. Ensure surge protection and backup power	5.1 Surge protection and backup power devices identified as per the requirement 5.2 Surge protection and backup power devices installed as per manufacturer's guidelines 5.3 Surge protection and backup power devices tested for proper functioning as per manufacturer's guidelines

Range Statement:

Work connected to this unit may take place at an office, workshop or any other appropriate location where limited Local Area Network (LAN) could be safely handled. Work related to this unit may be performed individually or as a member of team.

Elaboration of Terms in Performance Criteria

- **External devices in PC 4.1** may include: biometric devices using attendance and access control systems, network printers, surveillance system, cloud storage etc

Critical aspects

The assessment must confirm that the candidate is able to;

- Verify the functionality of network connectivity
- Verify the functionality of the internet & E- mail
- Verify the functionality of the external devices connected to the network

The following tools, equipment & material are included within this unit.

Tools and equipment	Materials
Computer system	Manuals
Operating system with bootable devices	Stationary
Device drivers	Network/Patch cables
Network devices	RJ-45 Jacks
Utility software	
Internet facility	
Application software	
PC repair tool box	
Network tool box	
Cable tester and probe	
Multimeter	
Drill	
PPE	

The following documents /References/Standards may be used for this unit:

- Software License
- Device manufacturers' guidelines
- Technical data sheet
- Standard Operating Procedure (SOP)
- Manufacturing warranty and terms and conditions
- Occupational Health and Safety guidelines

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Legal aspects of the computer software • Operating systems • Application software • Peripheral devices • Structured Cabling • Cabling standards • Fundamentals of networking • Functions of network devices • Communication techniques (guided (wired)/ unguided(wireless)) • Utility software • Ports, adapters and connectors • External devices which could be connected to computer network • Drivers and their application • Information security and safety • Operating system migration • System Cloning • Ethical use of computer software • Waste disposal methods	• Communication • Team work • Read and interpret manuals • Record keeping skills • Use of internet • Use of PPE • Cable laying and jointing

Worker behavior/Attitudes

Attitudes/Behavior
• Self confidence • Politeness • loyalty • Integrity and honesty • Punctual and efficient

Unit Title:	Practice workplace communication and interpersonal relations
Unit Descriptor:	This unit covers the competencies required to co-operate with others, interact effectively and deal with information within the workplace.
Unit Code:	K72S001BU01
Element	Performance Criteria
1. Co-operate with others in performing work	1.1 Work performed in co-operation with others in a friendly & courteous manner 1.2 Verbal and non-verbal communication used appropriately 1.3 Ideas, opinions, and interactions shared & respected appropriately 1.4 Work performed in co-operation with co-workers and related technical staff appropriately
2. Work with people of diverse backgrounds	2.1 People of a diverse backgrounds accepted without prejudice 2.2 Respect for cultural, ethnic and gender differences practised appropriately
3. Obtain record and convey workplace information	3.1 Effective questioning, active listening reading and speaking skills used to gather information 3.2 Information and ideas transferred without distortion 3.3 Work place interaction conducted in a courteous manner 3.4 Approved work instructions obtained and followed in appropriate manner 3.5 Collected information recorded for easy reference
4. Fill work related documents	4.1 Work related forms and formats identified appropriately 4.2 Range of forms and formats relating to work filled, appropriately 4.3 Reporting requirements to supervisor completed accurately in accordance with organizational guidelines and timelines.

Range/Context:

The performance of activities included in this unit may take place in workplace. Communication will include verbal, non-verbal, written, printed and electronic forms. It applies to individuals, groups and co-workers as well as others.

Critical Aspects:

The assessment must confirm that the candidate is able to.

- Obtain-accurate information from customer/co-workers
- Record essential information
- Convey essential information to relevant parties clearly and accurately

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment	Materials
• Calculator • Telephones • Audio – Video equipment	• Stationaries

The following documents /References/Standards may be used for this unit:

- Charts
- Graphics
- Memos
- Formats & Forms (Time Record sheets, Quality record sheets and material order sheets)

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Safety symbols & procedures • Mother tongue in both spoken and written form • Basic English relevant to workplace and type of work • Organization's rules, regulations, and procedures • General norms of polite conduct • Forms of address appropriate to customers and persons in different positions • Basic knowledge of customs and practices of different ethnic and religious groups • Good telephone manners • Basic understanding of gender issues • Basic knowledge of new trends in mobile phone technology	• Speak clearly in mother tongue and express varied ideas • Write clearly and legibly in mother tongue • Read, understand, and communicate simple English relevant to the workplace • Write simple explanations in English language • Communicate effectively using the telephone • Make simple sketches

Unit Title:	Apply occupational literacy and numeracy
Unit Descriptor:	This unit covers the competencies required to apply literacy and numeracy to participate effectively in activities in the workplace.
Unit Code:	K72S001BU02
Element	Performance Criteria
1. Apply basic mathematics to work tasks	1.1 Computation skills demonstrated using whole numbers, fractions, decimals and percentages manually or with calculators 1.2 Key concepts and principles of mathematics applied to understand and solve workplace calculations
2. Communicate ideas in writing	2.1 Adequate vocabulary maintained according to the nature of work/task 2.2 Documents filled and recorded to convey -the intended message 2.3 Appropriate units, codes and symbols selected according to purpose
3. Read range of written materials	3.1 Adequate vocabulary maintained to read and understand written materials according to workplace requirements 3.2 Appropriate units, codes and symbols read and interpreted for the task 3.3 Information gained from text, comprehend to communicate information to others

Range/Context:

Activities included in this unit may take place in workplace\ and will include but not be limited to

- Reading reports, instruction sheets, technical manuals and other types of written material in the mother tongue as well as in simple English
- Writing sentences in the mother tongue or in simple English
- Performing simple work-related calculations employing basic mathematical operations.

Critical Aspects:

The assessment must confirm that the candidate is able to;

- Interpret written material accurately
- Convey ideas clearly and correctly in writing & verbally
- Adopt appropriate method in mathematical calculations

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment	Materials
• Calculator • Audio- Video equipment	• Stationaries

The following documents /References/Standards may be used for this unit

- Charts
- Manuals
- Memos
- Formats & Forms (Time Record sheets, Quality record sheets and material order sheets)
- Accepted rules of grammar and syntax

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Fundamental principles of mathematics • Grammar, spelling, punctuation and syntax in mother tongue • Basic English grammar, spelling and syntax • Unit of measurement • Mathematical representation of physical quantities • Basic Mathematical operations • Adequate technical vocabulary in English • Principles of basic Geometry • Basic knowledge of new trends in the mobile phone technology	• Write clearly, legibly, and correctly in your mother tongue • Read and understand simple English relevant to the workplace • Write simple sentences in English • Perform simple calculations • Draw simple Geometrical views

Unit Title:	Work in teams
Unit Descriptor:	This unit covers the competencies required to work collaboratively and in cooperation with others and identify roles & responsibilities of members of the team.
Unit Code:	K72S001BU03
Element	Performance Criteria
1. Understand team role and scope	1.1 Formal and informal teams and their roles, goals and limitations identified in the workplace 1.2 Responsibilities, characteristics, and relationships of team members explained to accomplish team goals
2. Participate as a team member	2.1 Different forms of communication used with team members to contribute to team goals 2.2 Appropriate contribution (participation) made to the team to achieve team goals 2.3 Health and safety of team members ensured when performing team operations appropriately 2.4 Different viewpoints and ideas are respected as required to meet team goals 2.5 Collective decisions are accepted as required to meet team goals
3. Maintain productive group relations	3.1 Individual's role and responsibility within the team clearly identified 3.2 Other's roles and responsibilities identified and understood 3.3 Harmonious relations maintained and respected within the team and with others

Range/Context:

The performance of activities included in this unit may take place in workplace and will include but not be limited to identifying individual's and other's roles and responsibilities within the team and maintaining harmonious relations.

Critical Aspects:

The assessment must confirm that the candidate is able to;

- Achieve team goals and individual goals
- Follow designated work plan for the job
- Respect to other's ideas

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment	Materials
• Depending on the job assigned to the team for the purpose of assessing this unit, required tools, equipment and material will be provided.	• Stationaries

Documents/References/Standards

- Work plans
- Manuals
- Handbooks

Underpinning Knowledge and Skills

Underpinning Knowledge	Underpinning Skills
• Simple concepts of team behaviour • Elementary group dynamics • Basic knowledge in social and demographic structures	• Work effectively and comfortably as a member of a team • Patient • Understand others

Unit Title:	Practice occupational health and safety procedures in a workplace
Unit Descriptor:	This unit describes the performance outcomes required to identify and follow health, safety and emergency procedures in workplace. It involves those health and safety procedures to be followed when using workplace hand tools and power tools, fixed equipment, and chemicals.
Unit Code:	K72S001BU04
Element	Performance Criteria
1. Identify and follow workplace safety procedures	1.1 Workplace procedure relating to personal safety identified and followed 1.2 Unsafe situations and hazards/risks identified and reported according to workplace procedures 1.3 Machinery and equipment breakdowns identified and reported according to workshop procedures 1.4 Fire hazards identified 1.5 Different fire protection equipment and materials and their methods of use identified as required for different types of fires 1.6 Workplace procedures and safety requirements for handling and storing dangerous goods and hazardous substances identified and followed 1.7 Manual handling procedures and ergonomic parameters identified and followed 1.8 Health and Safety awareness programs participated 1.9 Appropriate PPE used/ worn according to the organisation's requirements & and manufacturer's instructions 1.10 First aid procedures performed as needed 1.11 Daily, weekly and periodical health & safety checklist followed including medical check-ups as per organization requirement
2. Identify and follow workplace emergency procedures	2.1 Potential mechanical, electrical, chemical, thermal and biological hazards recognized, identified and reported as per workplace procedures 2.2 Workplace emergency procedures regarding illness and accidents identified and followed 2.3 Safety warning alarms and workplace evacuation procedures, emergency protection area, fire exits, and fire extinguishers identified and followed 2.4 Firefighting appliances and equipment located and identified for emergency use and application 2.5 Relevant persons identified for contacting in the event of an incident, accident or sickness of self, co-workers, staff, customers and others 2.6 Accident and incident documentation requirements followed as per workplace procedures
3. Adhere to good environmental practices	3.1 Air, Water, Ground, Noise / Vibration, Energy identified to get the general overview of the environment

	3.2 Environmental pollution issues prevented to minimize the pollution related to the industry 3.3 Cleaner production concept followed 3.4 Waste disposal and segregation procedures followed 3.5 Environmental safety aspect followed to minimized the accident and acknowledge rules and regulations
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Range/Context:

Activities included in this unit may take place in a workplace and will include but not be limited to identifying potential hazards, taking effective preventive action to control or minimise such hazards, using personal protective equipment, using fire extinguishers and maintaining records of safety related activities.

Elaboration of terms:

Workplace procedures may include:

- Selecting and using personal protective equipment (PPE)
- Identifying hazards
- Emergency, fire and accident
- Regular maintenance of PPE & other machinery, tools and equipment
- Regular cleaning of Uniforms

First aid procedures may include but not be limited to treatment of minor cuts, bruises and burns, applying bandages & tourniquets, and neutralising the effects of corrosive substances.

Relevant persons must include:

Those holding a current recognized qualification in any of the following:

- First aid
- Cardiopulmonary resuscitation
- Fire warden and workplace evacuation
- Occupational or work health and safety representative.

Critical Aspects:

The assessment must confirm that the candidate is able to;

- Identify potential hazards
- Adhere to health & safety procedures
- Use most appropriate tools for a task

The following tools, equipment & material are included within this unit.

Tools/ instruments and equipment	Materials
• Safety signs and symbols • Colored/ barricaded tape and paint • Appropriate machine guards • Machine and safety devices • Safety shoes • Goggles • Gloves • Ear protectors • Eye wash fountain/shower • Safety belts • Helmets • First aid box with essential items • Fire protection equipment	• Posters • Stationaries

Documents/References/Standards

- Organisation's health & safety procedure manual
- Health & safety check list
- Factories' Ordinance No.45 of 1942 and its amendments
- Central Environmental Authority (CEA) rules and regulations related to industry (National Environment Act No.47 of 1980)
- MSDS
- Emergency contact details (Fire brigade, medical institute, police, insurance, etc)

Required Knowledge, Skills and Worker behavior/Attitude/Soft skills

Underpinning Knowledge	Underpinning Skills
Individuals must be able to demonstrate knowledge of: basic aspects of relevant OHS legislation, including: • Employer responsibilities for the workplace • Employee responsibilities to participate in health and safety practices • Employee responsibility to ensure own work practices protect the safety	• Read and understand valid test reports of plant and equipment • Read and understand valid test reports of Environmental aspects • Learning skills to: Locate sources of information relating to safety requirements. • Reading skills to: Interpret information from manufacturer and literature when seeking procedures required carrying out work safely.

of self, other workers and other people in the workplace roles of workplace personnel and personal safety in the workshop including: • Hearing protection • Skin protection • Protective clothing • Protective footwear • Eye protection manual handling, including: • Storing items • Availability of read and understand validity test • Planning the lift • Using correct lifting techniques and equipment workplace practices, including: • Housekeeping • Observing personal hygiene • Identifying slip hazards and trip hazards OHS symbols and signs, Machine, and equipment safety, including: • Using machine guards • Electrical tools, including: - Checking before use faulty equipment tagging and isolation procedures - Dangers of using electrical tools around water • Electrical, mechanical, chemical, thermal and biological hazards Emergency procedures, including: • Evacuation procedures and assembly points. • Personnel responsibilities Cleaner production methods	• Writing skills to: Legibly and accurately enter information into workplace safety documents and check sheets. • Oral communication skills to: Clarify instructions inform personnel of workplace hazards. • Numeracy skills to: Match hazardous material weights and volumes to comply with limitation given in Material Safety Data Sheets (MSDS) and work instructions. • Planning and organizing skills to: Achieve required outcome while maintaining health and safety regulations • Self-management skills to: Adhere to workplace safety on his own initiatives • Problem solving skills to: Recognize potential workplace hazards and initiate action Seek help from other sources in an uncertain situation to ensure the safety
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COMPETENCY BASED TRAINING CURRICULUM
FOR
COMPUTER HARDWARE TECHNICIAN
NVQ Level 04

List of Curriculum Modules		Module Code No	Duration (Hours)		Page
Section 2: Curriculum			Theory	Practical	
1.	Fundamentals of electricity and electronic	K72S001M01	18	18	39
2.	Computer fundamentals	K72S001M02	12	12	41
3.	Functions of Operating Systems, Utility Systems and Application Software	K72S001M03	18	24	42
4.	Components of the computer system	K72S001M04	36	30	43
5.	Entrepreneurship	K72S001M05	12	18	44
6.	Green IT Concept	K72S001M06	30	12	46
7.	Preparation of estimate/quotation	K72S001M07	18	18	48
8.	Fundamentals of cyber security	K72S001M08	12	06	49
9.	Troubleshooting and Repairing computer system	K72S001M09	18	36	50
10.	Assemble Computers	K72S001M10	12	48	52
11.	Maintain computers and peripherals	K72S001M11	18	36	53
12.	Install and configure computer software	K72S001M12	18	48	55
13.	Setup and Maintain Network systems	K72S001M13	30	48	57
14.	Communication skills for workplace	K72S001BM01	12	18	59
15.	Workshop calculation & Science	K72S001BM02	12	12	60
16.	Team work	K72S001BM03	12	18	61
17.	Occupational Safety & Health and Environmental Aspects	K72S001BM04	12	18	62
Total Duration			300	420	

Competency Profile

COMPUTER HARDWARE TECHNICIAN

COMPETENCY AREAS

COMPETENCIES

A. Gather customer requirement	A-1. Obtain customer requirement	A-2. Collect details of the solution	A-3. Suggest and agree solution with the customer	A-4. Record customer's details	A-5. Record visual inspections' details
	A-6. Obtain information from the customer regarding the fault	A-7. Complete job card			
B Prepare estimate/quotation	B-1. Select cost elements for the agreed solution/s	B-2. Prepare estimate/quotation for the agreed solution/s	B-3. Obtain approval of the prepared estimate/quotation	B-4. Submit prepared estimate/quotation to the customer	B-5. Obtain customer's approval for estimate/quotation
	B-6. Keep record on approved estimate/quotation				
C. Ensure the validity of warranty	C-1. Check warranty type of the computer system	C-2. Check condition/validity of the warranty seal before commencing the work	C-3. Inform to customer about the warranty terms and conditions		
D. Carry out preparation activity	D-1. Wear PPE/static protective cloths	D-2. Maintain personal hygiene	D-3. Maintain professional image	D-4. Collect/Purchase required components as per the order/job card	D-5. Verify collected/purchased components with the order/job card
E. Rectify issues of existing system	E-1. Obtain customer's concurrence for the detailed diagnosis of the computer system	E-2. Inspect computer system to localize the faulty area	E-3. Update job card	E-4. Diagnose Specific software/hardware/network fault of the inspected computer system and related peripherals	E-5. Select and arrange required hardware /network/software repair tools
	E-6. Clean Hardware/network components using cleaning agents	E-7. Attend hardware /network /software components	E-8. Check system operation to ensure the smooth function of the computer system		
F. Carry out post activity of the task	F-1. Dispose waste /E waste as per the environmental regulation	F-2. Clean used tools and equipment	F-3. Store cleaned tools and equipment	F-4. Arrange work bench	F-5. Review rectification work with the customer
	F-6. Obtain customer acceptance	F-7. Follow invoice and payment procedure			
G. Ensure the protection of computer system from cyber attack	G-1. Check activation/updating of antivirus	G-2. Ensure activation/updating of antivirus	G-3. Identify effect of malicious software by scanning	G4. Fix identified issues related to infections	G-5. Recheck functioning of computer software and devices
	G-6. Advise customer on protecting the computer system from cyber attack				

H. Perform assembling process	H-1. Arrange workplace as per the requirement	H-2. Assemble components	H-3. Ensure Completeness of assembling order/sequence of the assembled components	H-4. Connect power to the assembled computer system	H-5. Test system function using relevant diagnostic tools
	H-6. Forward assembled computer system for computer software installation				
I. Initiate the preventive maintenance work	I-1. Review preventive maintenance agreement	I-2. Gather required tools and equipment	I-3. Verify computer system as per the preventive maintenance system	I-04. Discuss current condition of the computer system with the user	I-05. Check current condition of the computer system
	I-06. Record current condition of the computer system	I-07. Create restore point	I-08. Carry out Backing up on the selected storage media		
J. Carry out hardware maintenance work	J-1. De assemble computer system	J-2. Dis charge static charges	J-3. De oxidize ports and connectors using contact cleaners	J-4. Replace thermal compound	J-5. Clean system case
	J-6. assemble computer system	J-7. Test assembled computer system for proper functioning	J-8. Restore back up		
K. Carry out Software maintenance work	K-1. Verify software issue/s	K-2. Record verified software issue/s	K-3. Update operating system	K-4. Clean temporary files	K-5. Perform disk cleanup
	K-6. Uninstall unnecessary /conflicting application software	K-7. Update device drivers	K-8. Update anti malware and other required utilities		
L. Instruct customer on trouble free operation	L-1. Inform current situation of Hardware/Software of the computer system to customer	L-2. Inform anticipated failures/Replacements of Hardware/Software of the computer system	L-3. Record anticipated failures/Replacements of Hardware/Software of the computer system	L-4. Describe and acknowledge Required settings of the computer system	
M. Install and configure operating system	M-1. Select operating system	M-2. Partition Hard Disk Drive (HDD)/Solid State Drive (SSD)	M-3. Format created partition	M-4. Perform installation of the selected operating system	M-5. Perform activation of the installed operating system
	M-6. Install/Update required device drivers	M-7. customize configuration of operating system	M-8. Test and verify functions for trouble free operations		
N. Install application software	N-1. Select application software as per the customer requirement	N-2. Verify compatibility of hardware & software for the selected application software	N-3. Customize infrastructure / software	N-4. Perform installation of the selected application software	N-5. Test and verify functions for trouble free operations
O. Install and configure utilities	O-1. select utility software as per the customer requirement	O-2. Verify compatibility of hardware & software for the selected utility software	O-3. Perform installation of the selected utility software	O-4. Customize configuration of the utility software	O-5. Test and verify functions for trouble free operations
P. Update software	P-1. Identify software update	P-2. Perform update of the selected software	P-3. Test and verify functions for trouble free operations	P-4. Make aware customer/user on updating procedures	

Q. Install hardware components	Q-1. Identify hardware components	Q-2. Check system capability for upgrading	Q-3. Verify system capability for upgrading	Q-4. List out required hardware components	Q-5. Fix obtained hardware components
	Q-6. Test and verify function for trouble free operation	Q-7. Keep record on install hardware components			
R. Setup external devices	R-1. Identify external devices/cloud storage	R-2. Verify compatibility of hardware / software of the external device/cloud storage	R-3. Install/configure External device/cloud storage and device drivers/software	R-4. Test and commission Functionality of the Installed device	R-5. Keep record on setup external devices
	R-6. Clean work environment				
S. Initiate the network requirement	S-1. Identify New/Upgrade computer network requirement	S-2. Identify solution	S-3. Discuss option with customer	S-4. Sketch Network layout and power supply layout	S-5. Obtain approval from the customer to finalized network layout
	S-6. Procure required item				
T. Setup / Install network	T-1. Perform physical connectivity	T-2. Install/Update required software	T-3. Configure computer network	T-4. Keep record on setup install network.	T-5. Clean work environment
U. Setup connectivity for communication	U-1. Configure internet connectivity	U-2. Configure Email account/s	U-3. Install/Configure chat tool/s	U-4. Configure social network setting	U-5. Keep record on setup connectivity for communication
V. Connect external devices to the network	V-1. Identify Connected/to be connected External devices	V-2. Install/Configure external devices	V-3. Test configured external devices		
W. Practice workplace communication and interpersonal relations	W.1 Co-operate with others in performing work	W.2 Work with people of diverse backgrounds	W.3 Obtain record and convey workplace information	W.4 Fill work related documents	
X. Apply occupational literacy and numeracy	X.1 Employ basic mathematics to complete work tasks and solve problems	X.2 Communicate ideas in writing	X.3 Read range of written materials		
Y. Work in teams	Y.1 Understand team role and scope	Y.2 Participate as a team member	Y.3 Maintain productive group relations		
Z. Practice occupational health and safety measures	Z.1 Follow health and safety rules/ procedures	Z.2 Minimize health or safety risks	Z.3 Maintain health and safety procedures	Z.4 Adhere to good environmental practices	

Programme Structure

Competency Area		Module		Competencies	Time	
		No.	Title		Institutional	
					Theory	Practical
	Basic Module (Technical)	K72S001M01	Fundamentals of electricity and electronic	Basic	18	18
		K72S001M02	Computer fundamentals		12	12
		K72S001M03	Functions of Operating Systems, Utility software and Application Software		18	24
		K72S001M04	Component of the computer system		36	30
		K72S001M05	Entrepreneurship		12	18
		K72S001M06	Green IT Concept		30	12
A	Gather customer requirement	K72S001M07	Preparation of estimate/quotation	A-1 to A-7	18	18
C	Ensure the validity of warranty			C-1 to C-3		
B	Prepare estimate/quotation			B-1 to B-6		
G	Ensure the protection of computer system from cyber attack	K72S001M08	Fundamentals of cyber security	G-1 to G-6	12	06
D	Carry out preparation activity	K72S001M09	Troubleshooting and Repairing computer system	D-1 to D-5	18	36
E	Rectify issues of existing system			E-1 to E-8		
F	Carry out post activity of the task			F-1 to F-7		
H	Perform assembling process	K72S001M10	Assemble Computers	H-1 to H-6	12	48
I	Initiate the preventive maintenance work	K72S001M11	Maintain computers and peripherals	I-1 to I-8	18	36
J	Carry out hardware maintenance work			J-1 to J-8		
K	Carry out Software maintenance work			K-1 to K-8		
L	Instruct customer on trouble free operation			L-1 to L-4		

Q	Install hardware components			Q-1 to Q-7		
R	Setup external devices			R-1 to R-6		
M	Install and configure operating system	K72S001M12	Install and Configure Computer Software	M-1 to M-8	18	48
N	Install application software			N-1 to N-5		
O	Install and configure utilities			O-1 to O-5		
P	Update software			P-1 to P-4		
S	Initiate the network requirement	K72S001M13	Setup and Maintain Network systems	S-1 to S-6	30	48
T	Setup / Install network			T-1 to T-5		
U	Setup connectivity for communication			U-1 to U-5		
V	Connect external devices to the network			V-1 to V-3		
W	Practice workplace communication and interpersonal relations	K72S001BM01	Communication skills for workplace	W-1 to W-4	12	18
X	Apply occupational literacy and numeracy	K72S001BM02	Workshop calculation & Science	X-1 to X-3	12	12
Y	Work in teams	K72S001BM03	Team Work	Y-1 to Y-3	12	18
Z	Practice occupational health and safety measures	K72S001BM04	Occupational Safety & Health and Environmental Aspects	Z-1 to Z-4	12	18

Mapping Table

Qualification	Relevant Units	Relevant Modules	Relevant tasks	Duration
K72S001Q1L3	U03 U04 U05 U06 BU01 BU02 BU03 BU04	M01 M02 M03 M04 M05 M06 M09 M10 M11 M12 BM01 BM02 BM03 BM04	D-1 to D-5 E-1 to E-8 F-1 to F-7 H-1 to H-6 I-1 to I-8 J-1 to J-8 K-1 to K-8 L-1 to L-4 Q-1 to Q-7 R-1 to R-6 M-1 to M-8 N-1 to N-5 O-1 to O-5 P-1 to P-4	588 Hrs.
K72S001Q2L4	U01 U02 U03 U04 U05 U06 U07 BU01 BU02 BU03 BU04	M01 M02 M03 M04vM05 M06 M07 M08 M09 M10 M11 M12 M13 BM01 BM02 BM03 BM04	A-1 to A-7 C-1 to C-3 B-1 to B-6 G-1 to G-6 D-1 to D-5 E-1 to E-8 F-1 to F-7 H-1 to H-6 I-1 to I-8 J-1 to J-8 K-1 to K-8 L-1 to L-4 Q-1 to Q-7 R-1 to R-6 M-1 to M-8 N-1 to N-5 O-1 to O-5 P-1 to P-4 S-1 to S-6 T-1 to T-5 U-1 to U-5 V-1 to V-3	720 Hrs.

Module Title	Fundamentals of electricity and electronic
Module Code	K72S001M01
Module Type	Basic (Technical)
Duration (Hrs.)	36 hour (T-18, P- 18)
Learning Outcomes	After completion of this module trainees will be able to • Explain the principles of electricity • Verify the functionality of the electronic components • Use testing tools like multimeter • Carry out Soldering/Desoldering work • Explain the importance of protecting the environment
Learning Content	Related Theory: • Principles of Electricity ○ Current, voltage, resistance ○ Ohm's law ○ Units & measurement ○ Electrical equations • Introduction to electronics • Electronic components ○ Resistors ○ Transistors ○ MOSFET ○ Thyristors ○ Capacitors ○ Diodes ○ Inductors ○ ICs • Introduction to fiber optic cables and optical transmission • Introduction to power supplies used in computer and related system • Heat sinking and thermal conducting methodology • Power protection devices (SPS, UPS, Surge arresters) • Noise filtering and shielding • Instruments used in electrical/electronic repairs ○ Soldering Iron / Hot gun ○ De soldering pump ○ Multimeter ○ Signal/Voltage generator ○ Oscilloscope ○ DC Power Supply • Behaviors of Electro Static Discharges (ESD) • Effects of Electro Magnetic Interferences (EMI) (Duration:18 hour)

	Practical: • Check AC, DC voltage and current, Resistance, Capacitance, Inductance using multimeter following safety procedures • Soldering/Desoldering selected components following safety procedures and environmental regulations (Duration :24 hour)
Teaching-Learning activities	• Illustrated talk • Demonstrations with Real objects • Video clips • Brainstorming session
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Computer fundamentals
Module Code	K72S001M02
Module Type	Basic (Technical)
Duration (Hrs.)	24 hours (T-12, P- 12)
Learning Outcomes	At the end of this module trainee should be able to; • Explain the basic functions of a computer system • Convert numbers between different number systems • Find required accurate information using relevant search engines • Send & receive Emails • Handle/use Social Network/ Media ethically
Learning Content	Theory: • Functions of the computer system ○ Input, Process, Output, Storage, Communication • Number systems ○ Binary, Octal, Decimal, Hexadecimal ○ Data representation methods (ASCII, Unicode) • Measuring storage capacities ○ bit, byte, kilobyte ... etc. • Types of data ○ Analog, Digital data • Use of Internet & Email ○ Use of Web browsers ○ Use of general search engines ○ Use of AI search engines ○ Work with Email • Use of Social Networks/Media (E.g. Facebook, LinkedIn, Google Classroom, YouTube, Instagram ... etc.) ○ Work with communities ○ Ethical use of Social Networks/Media ○ Legal aspects of computer software ○ Software license, activation (Duration:12 hour)
	Practical • Create an Email account • Send email to few peoples using Cc/Bcc options • Create a five-page report on a given topic by the authorized person • Present the created report on an audience (Duration:18 hour)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Functions of Operating Systems, Utility software and Application Software
Module Code	K72S001M03
Module Type	Basic (Technical)
Duration (Hrs.)	42 hour (T-18, P- 24)
Learning Outcomes	At the end of this module trainee should be able to; • Create folders/Files • Prepare a formatted document • Prepare a worksheet • Make a presentation using presentation software
Learning Content	Theory: • Introduction to computer operating systems • Introduction to mobile operating systems (Android, iOS, etc.) • Classification of operating systems ○ Single user/multi user ○ Commercial/ Free and Open-Source Software (FOSS) ○ Command Line Interface (CLI)/ Graphical User Interface (GUI) • Versions, editions and distributions of operating systems • Basic functions of an operating system ○ Handling Folders (Create, Rename, Delete, Move) ○ Setting up configuration (Date & time, regional settings, etc.) ○ Find files and folders • Basic setting of internet connectivity (Connection and Password) • Introduction to word processing software ○ Page setup (paper size, orientation, margins) ○ Formatting text (Bold, underline, italic, etc.) ○ Tables ○ Images • Introduction to spreadsheet software ○ Page setup (paper size, orientation, margins) ○ Basic functions • Introduction to presentation software (Duration-18 hour)
	Practical • Create a given folder structure • Create a curriculum vitae (CV) using word processing software • Prepare a quotation using spreadsheet software • Make a presentation for a selected topic in an audience (Duration-30 hour)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Component of the computer system
Module Code	K72S001M04
Module Type	Basic (Technical)
Duration (Hrs.)	66 hour (T36, P- 30)
Learning Outcomes	At the end of this module trainee should be able to; • Identify various components of a computer system • Explain the functions of each component • Check inter-compatibility of the selected component
Learning Content	Theory: • Introduction to different computer system ○ Personal computer/ Desktop ○ Work station and servers ○ Mobile computers/laptops • Components of a Computer system ○ Casing, power supply and power Adaptor ○ Motherboard ○ RAM/ROM ○ Processors (CPUs, Cooling system) ○ Adapter Cards and Expansion Slots ○ Hard Disk Drives and SSDs ○ Optical Storage Devices ○ Ports, Cables, and Adaptors • Peripheral devices ○ Input Devices ○ Output Devices (Monitors, Projectors, etc) ○ Storage devices ○ Communication devices (Duration - 36 hrs.) Practical • List the technical specification of the given component • Disassemble/ assemble components of a computer system following the relevant guidelines (Duration-42hrs.)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Entrepreneurship
Module Code	K72S001M05
Module Type	Basic module
Duration (Hrs.)	30 hour (T- 12, P- 18)
Learning Outcomes	After completion of this module the trainee will be able to: • Interpret entrepreneurship competencies and success factors • Describe Business launching formalities • Describe Financial administration tasks in a computer maintenance & repairing business • Perform customer awareness for computer maintenance & repairing business • Apply e-waste management aspects
Learning Content	Theory: • Introduction to entrepreneurship • Introduction to business plan • Business launching formalities ○ Business registration and licensing requirements ○ Business objectives, policies, procedures, strategies and technical guidelines ○ Legal, regulatory and ethical requirements • Financial tasks in a business ○ Basic income and expense budgets ○ Fixed and variable costs ○ Basic Account charts, weekly basis monthly basis – Cash book, ledger, etc. ○ Knowledge on Current market prices of material and accessories of appropriate quality ○ Banking procedure • Financial discipline and regular auditing/checking • Pricing structure for various tasks • Administration related functions: ○ Maintaining safe and healthy work environments ○ Maintaining Customer 's satisfaction at the reception counter – Pleasant appearance – Courtesy – Mannerism – Empathy • Purchasing procedure • Inventory controlling procedure ○ Maintaining inventory lists ○ Re-order procedure • HR aspects of team members ○ Keeping personal records ○ Statuary requirements

	○ Employee motivation ○ Mentoring /skill upgrading of team members ● Marketing approach ○ Goals and objectives ○ Target market ○ Branding for the repair center ○ Positive image built-up ○ Methods of publicity ● Green technology aspects ○ Electronic waste ○ E-waste disposal procedure (Duration: 12 Hour)
	Practical: ● Assignment-1: prepare a business plan on Entrepreneurship in computer maintenance & repairing business, Business launching formalities, financial tasks in a business, Administration factors, and Marketing approach ● Assignment-2: brief report on e- waste and disposal (Duration:18 hour)
Teaching-Learning activities	● Interactive lecture ● Group discussion ● Video presentation ● Assignments ● Demonstrations and practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Green IT Concept
Module Code	K72S001M06
Module Type	Basic
Duration (Hrs.)	42 hour (T- 30, P- 12)
Pre-Requisites	None
Learning Outcome/s	At the end of this module the student shall be able to: • Explain the principles and significance of Green IT • Evaluate the environmental and economic impact of IT systems • Implement methodologies to reduce energy consumption and e-waste in IT operations • Apply Green IT principles
Learning Content	• Green IT (06 hrs.) ○ What is Green IT? ▪ Definition and importance of Green IT ○ Environmental impact of IT ▪ E-waste, energy consumption, resource depletion, carbon footprint ▪ Paperless office concept ○ Regulatory frameworks and compliance ▪ ISO 14001: Environmental Management System ▪ WEEE Directive (Waste Electrical and Electronic Equipment Directive) ▪ LEED (Leadership in Energy and Environmental Design) for IT ▪ The Green Grid: Industry standards and best practices ▪ Energy Star ▪ E-waste disposal regulations ○ Benefits of adopting Green IT practices • Energy Efficiency in IT Infrastructure (12 hrs.) ○ Energy-efficient hardware, Software and components ▪ Selecting energy-efficient hardware components (processors, memory, storage) ▪ Implementing power management features in operating systems and devices ▪ Green software design and development practices ○ Power management techniques in data centers and servers ▪ Energy-intensive components in data centers (servers, storage, cooling) ○ Virtualization and consolidation for energy savings ▪ Benefits of virtualization for energy efficiency and resource optimization ▪ Server consolidation and virtualization technologies ○ Energy-efficient cooling solutions for IT facilities ▪ Energy-efficient cooling strategies (air cooling, water cooling) • Sustainable Software Development Practices (03 hrs) ○ Green software design and development practices ○ Green coding practices and tools • E-Waste Management and Recycling (06 hrs) ○ Types and composition of e-waste

	○ E-waste problem and its environmental and social impacts ○ E-waste recycling and disposal options ○ E-waste disposal regulations ○ Challenges in e-waste disposal ○ Strategies for reuse, recycling, and disposal ● Emerging Trends in Green IT (03 hrs) ○ Green Cloud Computing ○ Sustainable AI and machine learning ○ Renewable energy integration in IT systems ○ Internet of Things (IoT) for sustainability
	Practical 1: Field visit to an IT organization and prepare a report of Application of Green IT concept in that organization. Should consider the following factors when preparing the report: ● If no Green IT concept is used at present, should propose a Green IT system to them ● If they have a Green IT policy already, report should include proposals to improve their Green IT policy. Practical 2: Design an e-waste recycling plan for an IT organization Practical 3: Develop a Green IT strategy document for an IT organization Practical 4: Analysis of a real-world case studies of successful and unsuccessful (may be one each) Green IT initiatives
Teaching/Learning Activities	● Lectures ● Presentations ● Demonstration ● Practical ● Discussions using the material
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Preparation of estimate/quotation
Module Code	K72S001M07
Module Type	Core
Duration (Hrs.)	36 hour (T-18, P- 18)
Learning Outcomes	At the end of this module trainee should be able to; • Prepare estimate/quotation for new installation / repair work in the computer and peripheral devices / upgrading hardware & software / Preventive maintenance contract or agreement • Finalize the estimate/ quotation with the customer
Learning Content	Theory: • Introduction to costing and estimation principles • Collecting techniques of market data (specifications, prices, availability, validity, delivery, warranty terms) for new installation/repair work/maintenance work • Basic costing and estimation techniques ○ Material cost, Labour cost, Cost of fixed overhead, Traveling cost, Profit margins, Government taxes, Any other additional cost • Introduction to Required validity/timing period • Introduction to Compliances and conformities • Introduction to Supplier registration and tender procedures (Duration: 18 hour) Practical • Prepare an estimate/ quotation for the customer requirement • Negotiate and finalize the estimate quotation with the customer (Duration: 18 hour)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Fundamentals of cyber security
Module Code	K72S001M08
Module Type	Core
Duration (Hrs.)	18 hour (T-12, P- 06)
Learning Outcomes	At the end of this module trainee should be able to; • Scan the system for malware infections • Create a basic security policy document
Learning Content	Theory: • Introduction to cyber security • Malicious software (Malware) ○ Types of malwares ○ Degree of dangers from malware ○ Prevention methods of malware ○ Anti-malware • Introduction to network attacks • Software firewall • Introduction to social engineering attacks • Principles of security policy ○ User policy ○ Group policy (Duration -06 hour) Practical • Perform a system scan using a given anti-malware tool • Create a basic security policy document including user policy and group policy (Duration:18hour)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Troubleshooting and Repairing computer system
Module Code	K72S001M09
Module Type	Core
Duration (Hrs.)	54 hour (T-18, P- 36)
Learning Outcomes	At the end of this module trainee should be able to; • Identify Hardware & peripheral related problems • Identify Network & related devices problems • Identify OS and Application software issues and troubleshooting • Troubleshoot surveillance systems related issues
Learning Content	Theory: • Steps of troubleshooting ○ Troubleshooting flowcharts provided by the manufacturers • Common issues in computer components ○ Motherboard ○ Processor ○ RAM ○ VGA ○ Hard disk/SSD ○ Power Supply/Power Adaptor ○ Wireless Adaptor card ○ Sound Adaptor Card ○ Optical Drive ○ Keyboard ○ Touch pad ○ Interfaces and Ports • Common issues in peripheral devices ○ Monitors ○ Printers ○ Scanners ○ Projectors • Common issues in computer related additional devices ○ Biometric and access control devices ○ UPS/SPS ○ Routers/switches ○ Surveillance equipment • Common Operating System Related Issues ○ Startup/Boot Issues (BIOS) ○ Drivers ○ Fatal Error etc. ○ Upgrade/Update issues • Common application software related Issues ○ Compatibility issues ○ License issues • Importance of Ethical usage and Legal implications of unauthorized software/data usage (Duration- 18 hour)

	Practical • Perform troubleshooting following the instruction flowchart provided by the manufacturer • Perform troubleshooting for a given case (Duration: 48 hour)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Assemble Computers
Module Code	K72S001M10
Module Type	Core
Duration (Hrs.)	60 hour (T-12, P- 48)
Learning Outcomes	At the end of this module trainee should be able to; • Identify assembling tools • Assemble a complete personal computer • Verify assemble components in BIOS settings
Learning Content	Theory: • Tools used in assembling process • Assembling order for various components • Safety guidelines should follow in assembling process • BIOS and CMOS settings and configuration ○ POST ○ UEFI/Legacy configurations ○ Boot order settings • Fault Replicable Unit (FRUs) • Remove and replacement procedures (Duration: 12 hour)
	Practical • Assemble a complete personal computer following standard steps • BIOS setting configuration and saving • Verify the functionality of assembled components using utility software (Duration: 48 hour)
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Maintain computers and peripherals
Module Code	K72S001M11
Module Type	Core
Duration (Hrs.)	54 hour (T-18, P- 36)
Learning Outcomes	At the end of this module trainee should be able to; • Perform preventive maintenance of the computer system • Perform upgrade/update computer system • Troubleshoot security and maintenance issues
Learning Content	Theory: • Importance of computer maintenance • Benefits of Preventive Maintenance • Tools and equipment for computer maintenance • Safe Handling Practices ○ Electrostatic Discharge (ESD) awareness and prevention techniques ○ Proper handling of computer components and peripherals ○ Safety precautions while working inside a computer case • Assembling and disassembling a computer ○ Step-by-step guide to assembling a computer ○ Disassembling and reassembling computer components ○ Safety precautions and best practices • Booting a computer ○ BIOS/UEFI Configuration ○ Update the Firmware • Software Tools and Utilities ○ Disk cleanup ○ Defragmentation ○ Other related utility software • Security Policy ○ Importance of computer security and data protection ○ Identifying and avoiding malware threats (viruses, spyware, etc) ○ Password management and best practices ○ Basic firewall functionalities • Security Threats ○ Types of Malwares ○ Anti-Malware Programs ○ Remediating Infected Systems • Firewalls ○ Software Firewalls ○ Installing and Configuring Firewalls • Web Security ○ Browser Extensions and Plugins ○ Browser Privacy Settings ○ InPrivate Browsing ○ Pop-up Blocker • Protecting Data (Data Backup and Recovery)

	○ Restore point ○ Importance of data backup ○ Methods and tools for data recovery ○ Data Encryption ● Software Troubleshooting ○ Common software issues and solutions ○ Using system logs and error messages ○ Reinstalling and updating software ● Hardware Troubleshooting ○ Identifying hardware failures ○ Diagnostic tools and techniques ○ Replacing faulty components ● Network Troubleshooting ○ Common network issues and solutions ○ Tools for network diagnostics ○ Maintaining network security ○ User account ● Troubleshooting and maintaining peripheral devices ○ Printers ○ Scanners ○ UPS ○ Other devices ● Best Practices for Longevity ○ Power management and energy-saving tips ○ Environmental considerations ○ Documentation and record-keeping (Duration: 18 hour)
	Practical ● Perform preventive maintenance for the given hardware components ○ Safely de-assemble ○ Clean the components ● Perform upgrade/update computer system ● Troubleshoot security and maintenance issues (Duration: 48 hour)
Teaching-Learning activities	● Assignments ● illustrated talk ● Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Install and Configure Computer Software
Module Code	K72S001M12
Module Type	Core
Duration (Hrs.)	66 hour (T-18, P- 48)
Learning Outcomes	At the end of this module trainee should be able to; • Install operating systems and essential software applications • Configure software settings to optimize functionality for specific user needs • Troubleshoot basic software installation and configuration issues
Learning Content	Theory: • Importance of software installation and configuration • Introduction to Software ○ Types of software (operating systems, application software, utility software) ○ Software licensing models (commercial, open-source) ○ System requirements and compatibility considerations ○ Understanding software architecture and components • Operating System Installation ○ Different operating systems (Windows, macOS, Linux) ○ Planning for OS installation (backing up data, system requirements) ○ Installation process for chosen operating system (physical media or digital download) ○ User permissions and administrative rights • Other Software Installation (Application, Utility, etc.) ○ Software installation methods (download, installation media) ○ Software licenses and end-user license agreements (EULAs) ○ Software management practices (uninstalling software, updating applications) ○ Best practices for safe software downloads and avoiding malware • Software Updates ○ Importance of keeping software up-to-date ○ Configuring automatic updates ○ Manually updating the software • Software Licensing and Compliance ○ Types of software licenses (commercial, open-source, freeware) ○ Managing software licenses and subscriptions • System Maintenance Tools ○ Types of system utilities (cleanup tools, defragmentation, diagnostics) ○ Installing and configuring system maintenance tools ○ Regular system maintenance tasks • Software Configuration/Customization ○ Configuring software for specific user needs

	○ Customizing software settings for specific user needs ○ Customizing program interfaces and preferences ○ User accounts and permissions within software ● Installing and Configuring Security Software ○ Types of security software (antivirus, anti-malware, firewalls) ○ Installation and configuration of security applications ○ Running scans and configuring real-time protection (Duration: 18 hour)
	Practical ● Install given operating systems on the assembled computer ● Install given utility software ● Install given application software (Duration: 48 hour)
Teaching-Learning activities	● Assignments ● illustrated talk ● Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Setup and Maintain Network systems
Module Code	K72S001M13
Module Type	Core
Duration (Hrs.)	78 hour (T-30, P- 48)
Learning Outcomes	At the end of this module trainee should be able to; • Perform network cabling and configure network devices • Install and configure Network Operating Systems (NOS) • Configure network settings for IP addressing, subnet masks, and network protocols • Handle network resources like users, groups, and permissions. • Troubleshoot common network connectivity and performance issues • Apply security best practices for network protection
Learning Content	Theory: • Introduction to Networking ○ Importance of network systems ○ Network fundamentals (protocols, devices, media) ○ Network topologies (bus, star, mesh, etc.) ○ OSI and TCP/IP Model and its layers ○ Networking terminology and acronyms (IP address, subnet mask, gateway, etc.) • Network Cabling and Devices ○ Types of network cables ○ Cable crimping and termination techniques ○ Network devices (routers, switches, access points) ○ Functions of different network devices • Wireless Networking ○ Understanding wireless networking standards (Wi-Fi, Bluetooth) ○ Setting up wireless access points (WAPs) ○ Configuring wireless security (WPA2, WPA3) • IP Addressing and Subnetting ○ Understanding IP addressing (IPv4, IPv6) ○ Subnetting and subnet masks ○ Classful and classless addressing schemes ○ Private and public IP addresses ○ Calculating and assigning IP addresses • Installing Network Hardware ○ Installing and configuring network interface cards (NICs) ○ Setting up routers and switches ○ Connecting network cables and testing connections • Network Operating Systems ○ Installing and configuring network operating systems ○ Setting up network services (DHCP, DNS, FTP) ○ Managing users and permissions • Sharing of devices ○ File sharing

	○ Printer sharing • Network Security Fundamentals ○ Understanding network security principles ○ Implementing firewalls and intrusion detection systems (IDS) ○ Configuring VPNs and secure remote access • Network Monitoring and Management ○ Using network monitoring tools ○ Managing network performance and bandwidth ○ Troubleshooting network connectivity issues • Network Troubleshooting Techniques ○ Identifying and diagnosing network problems ○ Using diagnostic tools and commands (ping, traceroute, nslookup) ○ Resolving common network issues • Cloud services ○ Cloud storage services ○ Setup and configuration of cloud storage (Duration: 30 hour)
	Practical • Lay network cable, crimping keystone, connect to patch panel • Test network cable for connectivity • Setup network configuration for a given network • Create, configure and map cloud storage service
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Communication skills for workplace
Module Code	K72S001BM01
Module Type	Basic module
Duration (Hrs.)	30 hour (T-12, P-18)
Learning Outcomes	After completion of this module the trainee will be able to: - communicate ideas and information at the workplace effectively
Learning Content	Theory: • Listening skills • Telephone etiquettes • Presentation skills • Verbal and non-verbal communication • Safety symbols & procedures • Basic English relevant to workplace and type of work • Organization's rules, regulations and procedures • General norms of polite conduct • Forms of Different manner of address appropriately to customers, superior and subordinates and persons in different positions • Customs and practices of different ethnic and religious groups • Gender issues relevant to the communications • Trends in related technology • Friendliness through a friendly tone, a personal question, or simply a smile, encourage coworkers to engage in open and honest communication • Respect – convey respect for others and their ideas (Duration: 18 Hour)
	Practical: • Role plays and dramas • Assignments 01 - Provide opportunities to trainees to present learned content • Assignments 02 – Prepare list of 250 – 300 technical / vocational terms in English used in relevant industry during the course period • Assignment 03 – Collect manufacturers guidelines/ operational instructions / specifications in relevant field, prepare a file and interpret the information (Duration: 12 Hour)
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Workshop calculation & Science
Module Code	K72S001BM02
Module Type	Basic Module
Duration (Hrs.)	24 hour (T-12, P-12)
Learning Outcomes	After completion of this module the trainee will be able to: - perform basic arithmetic and trigonometric operations in the workplace when and where required. -Describe electrical and mechanical characteristics related to welding process -prepare development of surfaces by drawings
Learning Content	Theory • Basic Arithmetic • Area, Volume • Angles. • Units and Measurements (Length, Volume, Speed, Time) • Conversion from metric to BS and vice versa • Basic idea of Work, Energy, power & Pressure • Units & measurement: Description, care & use of :- Micrometers- Outside and depth micrometer, Micrometer adjustments, Vernier calipers, welding gauges
	Practical • Practice arithmetic and trigonometric operations, • Measuring Practice
Teaching-Learning activities	• Illustrated talk • Demonstrations • Group and individual practices • Use video clips • Assignments
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Team Work
Module Code	K72S001BM03
Module Type	Basic Module
Duration (Hrs.)	30 hours (T-12, P-18)
Learning Outcomes	At the end of this module student should be able to; • Contribute positively to the work in team environment • Work effectively with others in a socially diverse environment • Respect and understand the views of others • Give, receive and act upon feedback • Identify and describe own roles and roles of others
Learning Content	Theory: • Simple concepts of team behavior • Elementary group dynamics • Basic knowledge in social and demographic structures Practical • Identify and establish team purpose, team member's roles and responsibilities. • Coach and motivate team. • Assignment 01- Form a group for workshop cleaning, machine maintenance, waste management and give the feedback on group effort • Assignment 02- Arrange sports activities / social events
Teaching-Learning activities	• Assignments • Lecturers • role plays • case studies • team building activities • coaching and mentoring
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	Occupational Safety & Health and Environmental Aspects
Module Code	K72S001BM04
Module Type	Basic Module
Duration (Hrs.)	30 hours (T-12, P-18)
Learning Outcomes	After completion of this module the trainee will be able to: - describe importance of safety -use personal safety gears - arrange the place to minimize health risk - give first aid - Recognize and apply environmental aspects
Learning Content	Theory • Importance of Safety and general Precautions to be observed in the workshop. • safety signs - for danger, warning, caution & personal safety message • Safe handling of, Fire extinguishers used for Different types of fire. • Safety procedures - (General safety, Electrical safety, Chemical safety) • Types of waste materials • Environmental rules and regulations on disposal of e waste • Method of reuse of waste and scraps (Duration: 12 Hour) Practical • Practical related to Safety and Health, • Importance of maintenance and cleanliness of workplace • Demonstration on First aid and Fire safety, • Use personal safety gears • Arrange the place of work to minimize the health risks. • Arrange the place of work in order to make it possible to work in the most ergonomic way. • Give first aid. • Use of fire extinguishers • Demonstration on safe handling and Periodic testing of lifting equipment • Energy saving Tips • Collect details of waste (Duration: 18 Hour)
Teaching-Learning activities	• Illustrated talk • Use video clips • Demonstrations • Assignments
Forms of assessment	Formative Assessment + Summative Assessment

Assessment Guide:

Forms of assessment

Assessment shall be based on evidence collected through workplace performance or a combination of evidence collected through training and workplace performance.

Assessment context

This unit may be assessed on the job, off the job or a combination of on and off the job. The unit may be assessed individually.

Assessment conditions

The candidate will have access to:

- all tools, equipment, material and documentation required.

The candidate will be permitted to refer the following documents:

- material data sheets
- health and safety regulations
- product data sheet (PDS)
- company specification sheets
- manufacturer's instructional brochures and manuals

The candidate will be required to:

- orally or by other methods of communication, answer questions asked by the assessor.
- Identify superior relevant authorities and clients who can be approached for the collection of competency evidence where appropriate.
- present evidence of credit for any off-job training related to this unit.

Assessors must be satisfied that the candidate can competently and consistently perform all elements of the unit as specified by the criteria and that he/she possess the required underpinning knowledge.

Special notes

During assessment, the candidate will;

- Always demonstrate safe work practices
- Communicate information about processes, events or tasks being undertaken to ensure a safe and efficient working environment.
- Perform all tasks in accordance with standard operating procedures.
- perform all tasks to specifications
- take responsibility for the accuracy of his/her own work
- use accepted methods for calculation.

Tasks involved will be completed within reasonable time frames relating to typical workplace activities.

Resources required for assessment

These include material, tools and equipment listed within this unit

Tools, equipment and material required for the training program

(Approximately 20 students)

Power supply	05
Dc Soldering iron	10
Digital Multi meter / Analog multi meter	05
Magnified glass with light	05
Miniature / precisions screw driver set	05
Tweezers	25
Computer with accessories	04
Microscope (stereo type)	02
Work Table	10
Anti-static Brushers	05
Oxide cleaning pencil	05
Flat tools	05
Charging analyzer	04
USB cables	10
Thermal camera	01
Smart Screw holders	04
Hardware tool kit	02
Network tool kit	02
PPE kit	As required

NATIONAL COMPETENCY STANDARDS & COMPETENCY BASED TRAINING CURRICULUM For COMPUTER HARDWARE TECHNICIAN NVQ Level 03 and 04 (Competency Standard Code: K72S001)	
NATIONAL CERTIFICATE LEVEL 03 AND 04 IN THE OCCUPATION OF COMPUTER HARDWARE TECHNICIAN	
1. Endorsement date: 2025-02-06	2. Date of review: 2028-02-06
3. Purpose of the Qualification	To certify that the holder of this qualification has acquired the competencies contained in the units listed in section 5 below.
4. Regulations for the Qualification/s	The holder should have been assessed by a panel of assessor and found competent in the units listed in section 5 and approved by the TVEC
5. Qualification Code & Packages	• U03+ U04+ U05 + U06+BU01+ BU02+ BU03+ BU04 – K72S001Q1L3 (Assistant Computer Hardware Technician) • U01+ U02+ U03+ U04 + U05+ U06 + U07+ BU01+ BU02+ BU03+ BU04 – K72S001Q2L4 (Computer Hardware Technician)
6. Prerequisite	• For NVQ Level 03 Qualification – None • For NVQ Level 04 Qualification – Passed GCE(O/L) Examination or having relevant NVQ Level 3 qualification
7. Accreditation Requirement	The qualifications shall be offered in compliance with the accreditation requirements of the TVEC as stipulated in the National Vocational Qualifications framework of Sri Lanka.
8. Certification	TVEC shall certify the qualifications in terms of the regulation at section 5 above.
9. Transition Arrangements	The competency-based assessments shall be undertaken by the registered assessors of the TVEC.
10. Requirement of conducting Knowledge Assessment	Not Required
11. Contact for Comments	Chairman, National Apprentice & Industrial Training Authority, 971, Sri Jayawardanapura Mawatha, Welikada, Rajagiriya. Director General, Tertiary & Vocational Education Commission, 354/2, Nipunatha Piyasa, Elvitigala Mawatha, Colombo 05

ACKNOWLEDGEMENT (ORIGINAL DEVELOPMENT)

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